ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2025 P2HR Worked Solutions

Jun 2025 | P2HR

33 questions ▪ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

TOPIC: **Solving Linear Equations**

Jun 2025 P2HR - Jun 2025 | P2HR

Q#: 1 Marks: 5

1 (a) Factorise fully $18c - 45cd$

....

(2)

(b) Solve $\dfrac{5 - 2x}{6} = 3x - 4$

Show clear algebraic working.

$x =$ ....

(3)

(Total for Question 1 is 5 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 5

TOPIC: **Solving Linear Equations**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Corrected to solving linear equations. The original tag was factorising, but the equation is the larger part.

Solution

(a)

$$18c - 45cd = 9c(2 - 5d)$$

(b)

$$\frac{5 - 2x}{6} = 3x - 4$$

$$5 - 2x = 18x - 24$$

$$29 = 20x$$

$$x = \frac{29}{20}$$

FINAL ANSWER

(a) $9c(2 - 5d)$, (b) $x = \frac{29}{20}$

PAST PAPER QUESTION

Q#: 2 Marks: 3

Topic: **Prime Factors, HCF & LCM**

Jun 2025 P2HR - Jun 2025 | P2HR

2 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$1400 = 14 \times 100$$

$$1400 = (2 \times 7)(2^2 \times 5^2)$$

$$1400 = 2^3 \times 5^2 \times 7$$

FINAL ANSWER

$2^3 \times 5^2 \times 7$

PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Simultaneous Equations**

Jun 2025 P2HR - Jun 2025 | P2HR

3 Solve the simultaneous equations

$$3x + 2y = 10$$
$$3x - 4y = 16$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Simultaneous Equations**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Simultaneous equations. The tag is correct.

Solution

$$3x + 2y = 10$$

$$3x - 4y = 16$$

Subtract the second equation from the first:

$$6y = -6$$

$$y = -1$$

Substitute into $3x + 2y = 10$:

$$3x + 2(-1) = 10$$

$$3x = 12$$

$$x = 4$$

FINAL ANSWER

$x = 4$, $y = -1$.

PAST PAPER QUESTION

Q#: 4 **Marks: 5**

TOPIC: **Percentages**

Jun 2025 P2HR - Jun 2025 | P2HR

4 Joshua is going to cover a floor with tiles for a customer.
The area of the floor is 45 m^2

Joshua buys one box of tiles for each 1.5 m^2 of floor area.
Each box of tiles costs £64

Joshua also buys 5 bags of tile adhesive.
Each bag of tile adhesive costs £12

Joshua charges the customer £3000

Work out his percentage profit.
Give your answer correct to one decimal place.

.... %

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 5

TOPIC: **Percentages**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Percentages. The tag is correct because the final task is percentage profit.

Method

Number of boxes of tiles:

$$\frac{45}{1.5} = 30$$

Cost of tiles:

$$30 \times 64 = 1920$$

Cost of adhesive:

$$5 \times 12 = 60$$

Total cost:

$$1920 + 60 = 1980$$

Profit:

$$3000 - 1980 = 1020$$

Percentage profit:

$$\frac{1020}{1980} \times 100 = 51.515\ldots\%$$

FINAL ANSWER

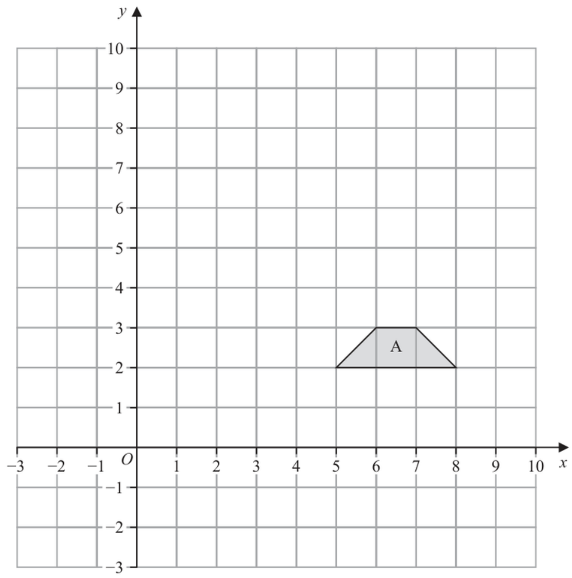
51.5%

PAST PAPER QUESTION Q#: 5 Marks: 4

TOPIC: **Transformations**

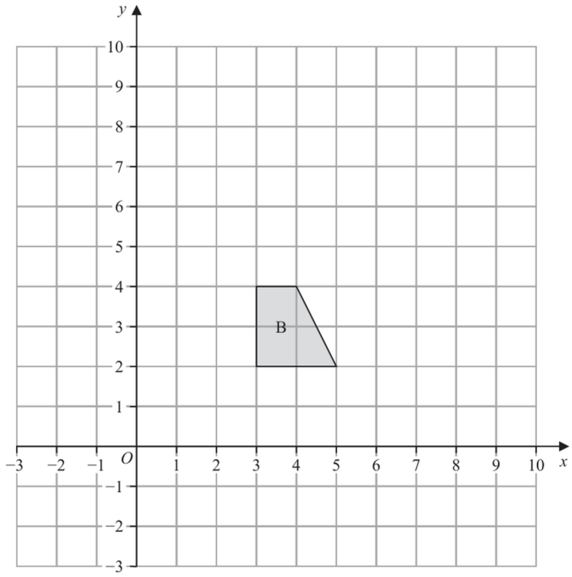
Jun 2025 P2HR - Jun 2025 | P2HR

5



(a) On the grid above, reflect shape **A** in the line $y = x$

(2)



(b) On the grid above, enlarge shape **B** by scale factor 2 with centre (1, 1)

(2)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Transformations**

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WORKED SOLUTION

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Topic check. Transformations. The tag is correct.

Solution

(a) Reflection in $y = x$ swaps each coordinate:

$$(x, y) \mapsto (y, x)$$

The image of A has vertices

$$(2, 5),\ (2, 8),\ (3, 7),\ (3, 6)$$

(b) Enlarge B by scale factor 2 from centre $(1, 1)$.
The vertices become

$$(5, 3),\ (9, 3),\ (7, 7),\ (5, 7)$$

FINAL ANSWER

(a) vertices $(2, 5), (2, 8), (3, 7), (3, 6)$; (b) vertices $(5, 3), (9, 3), (7, 7), (5, 7)$.

PAST PAPER QUESTION

Q#: 6 Marks: 5

TOPIC: **Algebraic Roots & Indices**

Jun 2025 P2HR - Jun 2025 | P2HR

6 (a) Write down the value of 5^0

....
(1)

$$\frac{5^9 \times 5^{-3}}{5^{-2}} = 5^k$$

(b) Find the value of k

k = ...
(2)

(c) Simplify fully $\left(2d^4e^5\right)^3$

....
(2)

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 5

TOPIC: **Algebraic Roots & Indices**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Algebraic Roots and Indices. This question uses index laws, not proportion.

Method

$$5^0 = 1$$

For part (b),

$$\frac{5^9 \times 5^{-3}}{5^{-2}} = 5^{9+(-3)-(-2)} = 5^8$$

So

$$k = 8$$

For part (c),

$$(2d^4e^5)^3 = 2^3d^{4\times3}e^{5\times3}$$

$$= 8d^{12}e^{15}$$

FINAL ANSWER

(a) 1, (b) $k = 8$, (c) $8d^{12}e^{15}$.

PAST PAPER QUESTION

TOPIC: **Standard & Compound Units**

Jun 2025 P2HR - Jun 2025 | P2HR

Q#: 7 Marks: 2

7 The mass of a silver coin is 48.3 g
The density of silver is 10.5 g/cm^3

Work out the volume of the silver coin.

.... cm^3

(Total for Question 7 is 2 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 2

TOPIC: **Standard & Compound Units**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Standard & Compound Units. The tag is correct.

Solution

Use

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

So

$$\text{volume} = \frac{\text{mass}}{\text{density}} = \frac{48.3}{10.5} = 4.6$$

FINAL ANSWER

$4.6\ \text{cm}^3$.

PAST PAPER QUESTION **Q#: 8** **Marks: 3**

TOPIC: **Statistics Toolkit**

Jun 2025 P2HR - Jun 2025 | P2HR

8 The mean of 7 numbers is 60

The mean of 3 of the numbers is 46

Work out the mean of the other 4 numbers.

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The total of the 7 numbers is

$$7 \times 60 = 420$$

The total of the 3 numbers is

$$3 \times 46 = 138$$

The total of the other 4 numbers is

$$420 - 138 = 282$$

Their mean is

$$\frac{282}{4} = 70.5$$

FINAL ANSWER

70.5.

PAST PAPER QUESTION Q#: 9 Marks: 3

TOPIC: **Percentages**

Jun 2025 P2HR - Jun 2025 | P2HR

9 In a sale, normal prices are reduced by 15%

The sale price of a dishwasher is 612 Swiss francs.

Work out the normal price of the dishwasher.

.... Swiss francs

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Percentages**

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WORKED SOLUTION

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Topic check. Percentages. The tag is correct.

Method

A reduction of 15% means the sale price is 85% of the normal price.

$$0.85 \times \text{normal price} = 612$$

$$\text{normal price} = \frac{612}{0.85} = 720$$

FINAL ANSWER

720 Swiss francs

PAST PAPER QUESTION

Q#: 10 **Marks: 2**

TOPIC: **Linear Graphs ($y = mx + c$)**

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10 A straight line, **L**, is parallel to the line with equation $y = 2 - 5x$
The line **L** passes through the point (0, 6)

Find an equation of the line **L**

(Total for Question 10 is 2 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 2

TOPIC: **Linear Graphs (y = mx + c)**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Correct. This is a parallel straight-line equation question.

Solution

The line

$$y = 2 - 5x$$

has gradient -5.
A parallel line has the same gradient, so line L has form

$$y = -5x + c$$

Line L passes through $(0, 6)$, so

$$6 = c$$

FINAL ANSWER

$y = -5x + 6.$

PAST PAPER QUESTION Q#: 11 Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P2HR - Jun 2025 | P2HR

11 The diagram shows two triangles, *ADE* and *CDB*

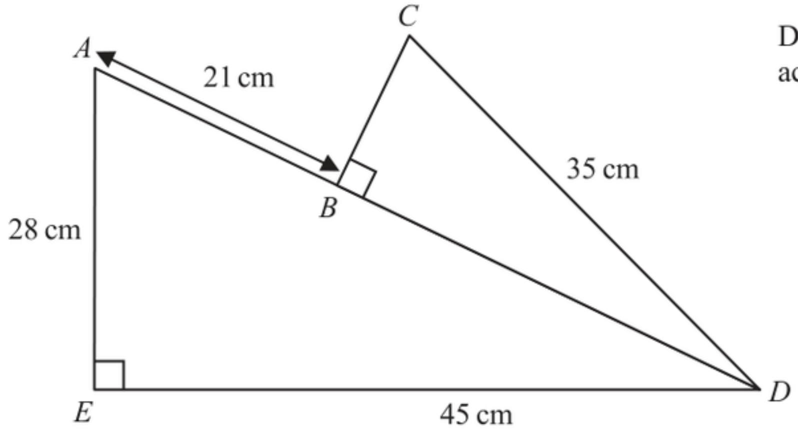

Diagram **NOT** accurately drawn

ABD is a straight line.

$AE = 28\,\text{cm}$ $ED = 45\,\text{cm}$ $AB = 21\,\text{cm}$ $CD = 35\,\text{cm}$

angle AED = angle $CBD = 90°$

Work out the area of triangle *CDB*
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Corrected to right-angled triangles. The solution uses Pythagoras twice.

Solution

In right-angled triangle AFD,

$$AD^2 = 28^2 + 45^2$$

$$AD^2 = 2809$$

$$AD = 53$$

Since $AB = 21$,

$$BD = 53 - 21 = 32$$

In right-angled triangle CBD,

$$CB^2 = 35^2 - 32^2$$

$$CB^2 = 201$$

$$CB = \sqrt{201}$$

Area of triangle CDB is

$$\frac{1}{2} \times BD \times CB$$

$$\text{Area} = \frac{1}{2} \times 32 \times \sqrt{201} = 226.9\ldots$$

FINAL ANSWER

227 cm^2 to 3 significant figures.

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P2HR - Jun 2025 | P2HR

12 Osman buys a car for \$16 000

The car depreciates at a rate of 12% each year for the first 2 years.
In the third year, the car depreciates at a rate of x%

At the end of 3 years, the value of the car is \$11 461.12

Work out the value of x

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

After the first 2 years,

$$16000(0.88)^2 = 12390.40$$

In the third year, the value becomes 11461.12, so

$$12390.40\left(1 - \frac{x}{100}\right) = 11461.12$$

$$1 - \frac{x}{100} = \frac{11461.12}{12390.40} = 0.925$$

$$x = 7.5$$

FINAL ANSWER

$x = 7.5$

PAST PAPER QUESTION Q#: 13 Marks: 5

TOPIC: **Expanding Brackets**

Jun 2025 P2HR - Jun 2025 | P2HR

13 (a) Factorise $4x^2 - 25y^2$

....

(2)

(b) Show that $4x(x+3)(2x-5)$ can be written in the form $ax^3 + bx^2 + cx$ where a, b and c are integers to be found.

(3)

(Total for Question 13 is 5 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 5

TOPIC: **Expanding Brackets**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to expanding brackets. The original tag was factorising, but the expansion part carries more marks.

Solution

(a)

$$4x^2 - 25y^2 = (2x - 5y)(2x + 5y)$$

(b)

$$4x(x + 3)(2x - 5)$$

First,

$$(x + 3)(2x - 5) = 2x^2 + x - 15$$

Then

$$4x(2x^2 + x - 15) = 8x^3 + 4x^2 - 60x$$

So $a = 8$, $b = 4$, and $c = -60$.

FINAL ANSWER

(a) $(2x - 5y)(2x + 5y)$, (b) $8x^3 + 4x^2 - 60x$

PAST PAPER QUESTION Q#: 14 Marks: 7

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

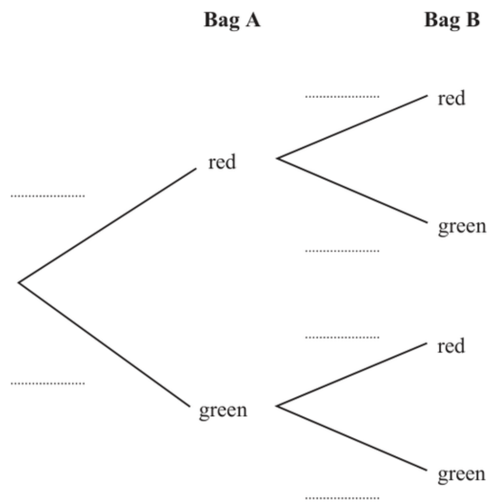
Jun 2025 P2HR - Jun 2025 | P2HR

14 Sara has two bags, **A** and **B**

In bag **A**, there are only 5 red beads and 4 green beads.
In bag **B**, there are only 7 red beads and 3 green beads.

Sara takes at random a bead from bag **A**
She then takes at random a bead from bag **B**

(a) Use this information to complete the probability tree diagram.



(2)

(b) Work out the probability that Sara takes two red beads.

....

(2)

Sara puts the beads back into their original bags.

Sara also has a box of beads.
In the box, there are only red beads and green beads.

When a bead is taken at random from the box, the probability that it is a green bead is $\frac{2}{11}$

Sara takes at random a bead from bag **A**
She then takes at random a bead from bag **B**
She then takes at random a bead from the box.

(c) Work out the probability that Sara takes more red beads than green beads.

....

(3)

(Total for Question 14 is 7 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 7

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

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WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

Bag A has 5 red and 4 green beads:

$$P(A_R) = \frac{5}{9}, \qquad P(A_G) = \frac{4}{9}$$

Bag B has 7 red and 3 green beads:

$$P(B_R) = \frac{7}{10}, \qquad P(B_G) = \frac{3}{10}$$

Two red beads:

$$\frac{5}{9} \times \frac{7}{10} = \frac{7}{18}$$

For part (c), the box has

$$P(\text{green}) = \frac{2}{11}, \qquad P(\text{red}) = \frac{9}{11}$$

More red than green means at least two red beads:

$$RRR + RRG + RGR + GRR$$

$$\frac{5}{9}\frac{7}{10}\frac{9}{11} + \frac{5}{9}\frac{7}{10}\frac{2}{11} + \frac{5}{9}\frac{3}{10}\frac{9}{11} + \frac{4}{9}\frac{7}{10}\frac{9}{11} = \frac{386}{495}$$

FINAL ANSWER

$\frac{7}{18}$, $\frac{386}{495}$.

PAST PAPER QUESTION

Q#: 15 **Marks: 3**

TOPIC: **Circle Theorems**

Jun 2025 P2HR - Jun 2025 | P2HR

15 A, B and C are points on a circle, centre O

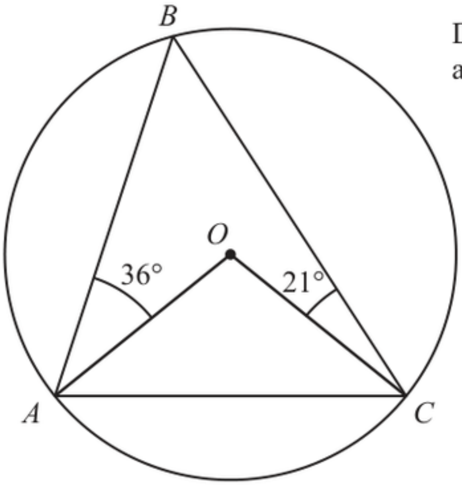

Diagram **NOT** accurately drawn

Angle $BAO = 36°$
Angle $BCO = 21°$

Work out the size of angle ACO

.... °

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Circle Theorems**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Circle Theorems. The tag is correct.

Solution

Since $OA = OB$, triangle AOB is isosceles:

$$\angle ABO = 36^\circ$$

So

$$\angle AOB = 180^\circ - 36^\circ - 36^\circ = 108^\circ$$

Since $OB = OC$, triangle BOC is isosceles:

$$\angle OBC = 21^\circ$$

So

$$\angle BOC = 180^\circ - 21^\circ - 21^\circ = 138^\circ$$

Angles around O add to 360°:

$$\angle AOC = 360^\circ - 108^\circ - 138^\circ = 114^\circ$$

Since $OA = OC$,

$$\angle ACO = \frac{180^\circ - 114^\circ}{2} = 33^\circ$$

FINAL ANSWER

33°.

PAST PAPER QUESTION

Q#: 16 Marks: 3

Topic: **Algebraic Proof**

Jun 2025 P2HR - Jun 2025 | P2HR

16 Show that $\dfrac{4}{3\sqrt{5}+7}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show each stage of your working.

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is acceptable because the question asks for exact-form proof.

Solution

$$\frac{4}{3\sqrt{5}+7} \times \frac{7-3\sqrt{5}}{7-3\sqrt{5}} = \frac{4(7-3\sqrt{5})}{49-45}$$

$$= 7-3\sqrt{5}$$

Since $3\sqrt{5} = \sqrt{45}$,

$$7-3\sqrt{5} = 7-\sqrt{45}$$

FINAL ANSWER

$7-\sqrt{45}$, so $a = 7$ and $b = 45$.

PAST PAPER QUESTION

TOPIC: **Rearranging Formulas**

Jun 2025 P2HR - Jun 2025 | P2HR

Q#: 17 Marks: 4

17 Make k the subject of $p = \dfrac{8k^2 + 5}{7 - 3k^2}$

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Rearranging Formulas**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Rearranging formulas. The tag is correct.

Solution

$$p = \frac{8k^2 + 5}{7 - 3k^2}$$

$$p(7 - 3k^2) = 8k^2 + 5$$

$$7p - 5 = k^2(3p + 8)$$

$$k^2 = \frac{7p - 5}{3p + 8}$$

FINAL ANSWER

$k = \pm\sqrt{\frac{7p-5}{3p+8}}$

PAST PAPER QUESTION

Q#: 18 Marks: 3

TOPIC: Vectors

Jun 2025 P2HR - Jun 2025 | P2HR

18 OAB is a triangle.

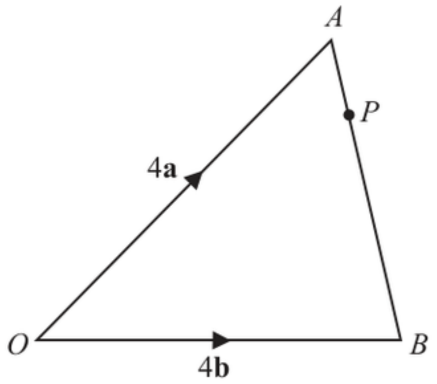

Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 4\mathbf{a}$

$\overrightarrow{OB} = 4\mathbf{b}$

P is the point on AB such that $AP : PB = 1 : 3$

(a) Write down $\overrightarrow{AB}$ in terms of **a** and **b**

....

(1)

(b) Express $\overrightarrow{OP}$ in terms of **a** and **b**
Give your answer in its simplest form.

....

(2)

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Vectors**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Corrected to Vectors.

Solution

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4\mathbf{b} - 4\mathbf{a}$$

Since $AP : PB = 1 : 3$, point P is one quarter of the way from A to B.

$$\overrightarrow{AP} = \frac{1}{4}\overrightarrow{AB} = \frac{1}{4}(4\mathbf{b} - 4\mathbf{a}) = \mathbf{b} - \mathbf{a}$$

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 4\mathbf{a} + \mathbf{b} - \mathbf{a} = 3\mathbf{a} + \mathbf{b}$$

FINAL ANSWER

$\overrightarrow{AB} = 4\mathbf{b} - 4\mathbf{a}$, and $\overrightarrow{OP} = 3\mathbf{a} + \mathbf{b}$.

PAST PAPER QUESTION Q#: 19 Marks: 3

TOPIC: **Functions**

Jun 2025 P2HR - Jun 2025 | P2HR

19 The functions f and g are such that

$$f(x) = 3x - 4 \quad \text{where } x > 2$$

$$g(x) = \frac{x}{2x + 1}$$

(a) State the value of x that cannot be included in any domain of g

....

(1)

(b) Find gf(x)
Give your answer in its simplest form.

gf(x) = ...

(2)

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Functions**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Functions. The tag is correct.

Solution

$$f(x) = 3x - 4, \qquad g(x) = \frac{x}{2x+1}$$

(a) The denominator of g cannot be zero:

$$2x + 1 = 0$$

$$x = -\frac{1}{2}$$

(b)

$$gf(x) = g(f(x))$$

$$gf(x) = \frac{3x-4}{2(3x-4)+1}$$

$$gf(x) = \frac{3x-4}{6x-7}$$

FINAL ANSWER

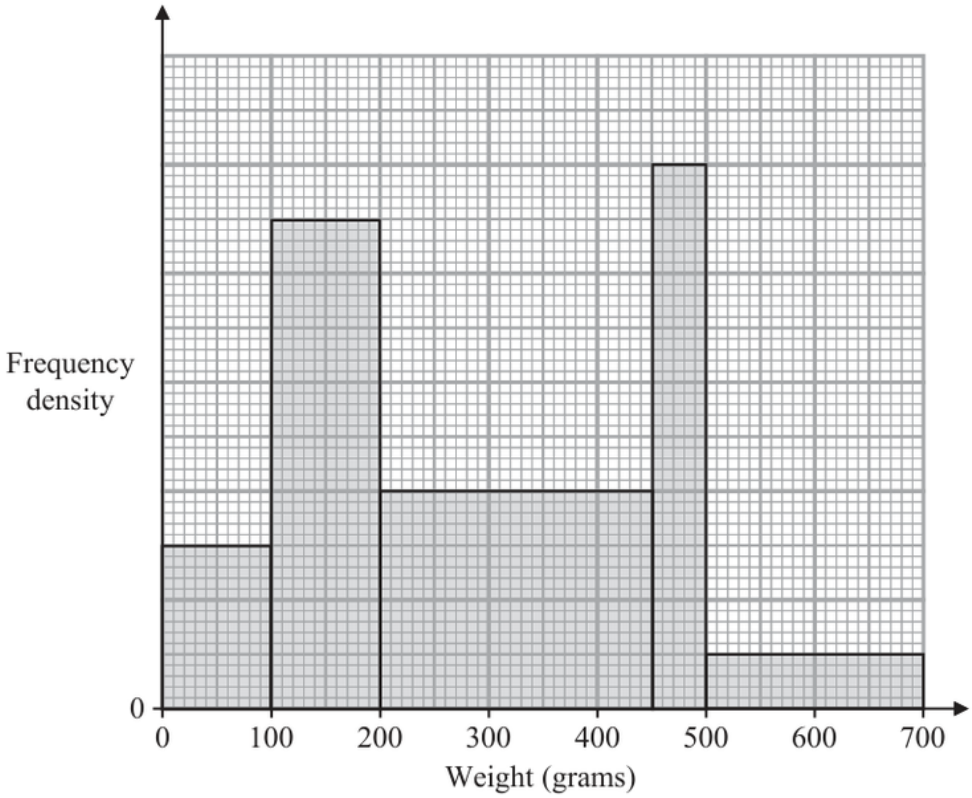
Excluded value $-\frac{1}{2}$, and $gf(x) = \dfrac{3x-4}{6x-7}$.

PAST PAPER QUESTION **Q#: 20** **Marks: 4**

TOPIC: **Histograms**

Jun 2025 P2HR - Jun 2025 | P2HR

20 The histogram gives some information about the weights, in grams, of some books.



75 books weigh less than 100 grams.

A book is chosen at random.

Find an estimate for the probability that this book weighs between 400 grams and 600 grams.

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

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Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For books weighing less than 100 g, the frequency is 75 and the class width is 100, so

$$\text{frequency density} = \frac{75}{100} = 0.75$$

Using this scale from the histogram, the frequency densities are:

$$0.75,\ 2.25,\ 1.0,\ 2.5,\ 0.25$$

The total number of books is

$$(100)(0.75) + (100)(2.25) + (250)(1.0) + (50)(2.5) + (200)(0.25)$$

$$= 75 + 225 + 250 + 125 + 50 = 725$$

For weights between 400 g and 600 g:

$$(50)(1.0) + (50)(2.5) + (100)(0.25)$$

$$= 50 + 125 + 25 = 200$$

So the probability is

$$\frac{200}{725} = \frac{8}{29}$$

FINAL ANSWER

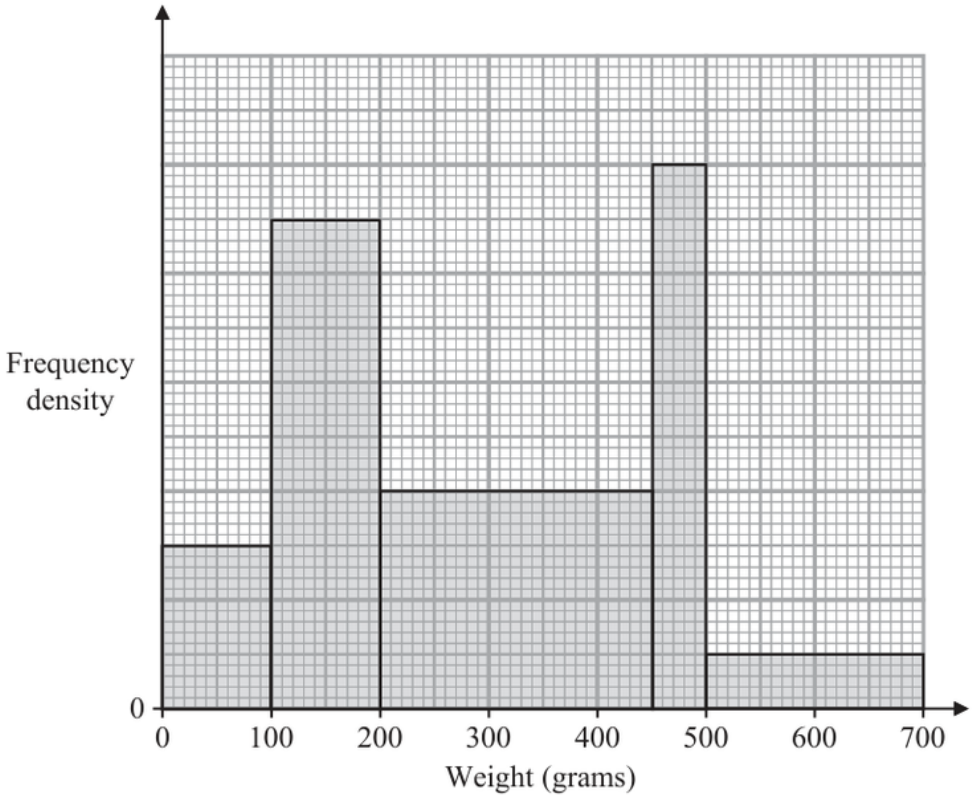
$\frac{8}{29}$ or 0.276 to 3 significant figures

PAST PAPER QUESTION **Q#: 20** **Marks: 4**

TOPIC: **Histograms**

Jun 2025 P2HR - Jun 2025 | P2HR

20 The histogram gives some information about the weights, in grams, of some books.



75 books weigh less than 100 grams.

A book is chosen at random.

Find an estimate for the probability that this book weighs between 400 grams and 600 grams.

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For books weighing less than 100 g, the frequency is 75 and the class width is 100, so

$$\text{frequency density} = \frac{75}{100} = 0.75$$

Using this scale from the histogram, the frequency densities are:

$$0.75,\ 2.25,\ 1.0,\ 2.5,\ 0.25$$

The total number of books is

$$(100)(0.75) + (100)(2.25) + (250)(1.0) + (50)(2.5) + (200)(0.25)$$

$$= 75 + 225 + 250 + 125 + 50 = 725$$

For weights between 400 g and 600 g:

$$(50)(1.0) + (50)(2.5) + (100)(0.25)$$

$$= 50 + 125 + 25 = 200$$

So the probability is

$$\frac{200}{725} = \frac{8}{29}$$

FINAL ANSWER

$\frac{8}{29}$ or 0.276 to 3 significant figures

PAST PAPER QUESTION Q#: 21 Marks: 3

TOPIC: **Solving Inequalities**

Jun 2025 P2HR - Jun 2025 | P2HR

21 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Solving Inequalities**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$2x^2 - 7x - 15 > 0$$

Factorise:

$$2x^2 - 7x - 15 = (2x + 3)(x - 5)$$

The critical values are

$$x = -\frac{3}{2} \quad \text{and} \quad x = 5$$

The quadratic is greater than 0 outside the roots.

FINAL ANSWER

$x < -\frac{3}{2}$ or $x > 5$

PAST PAPER QUESTION

Q#: 21 Marks: 3

TOPIC: **Solving Inequalities**

Jun 2025 P2HR - Jun 2025 | P2HR

21 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Solving Inequalities**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$2x^2 - 7x - 15 > 0$$

Factorise:

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The quadratic is greater than 0 outside the roots.

FINAL ANSWER

$x < -\frac{3}{2}$ or $x > 5$

PAST PAPER QUESTION Q#: 22 Marks: 5

TOPIC: **Differentiation**

Jun 2025 P2HR - Jun 2025 | P2HR

22 The diagram shows a solid cuboid.

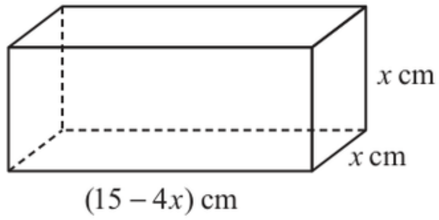

Diagram **NOT** accurately drawn

The volume of the cuboid is V cm^3

Find the maximum value of V

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Differentiation**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The cuboid has dimensions

$$x, \quad x, \quad 15 - 4x$$

So

$$V = x^2(15 - 4x)$$

$$V = 15x^2 - 4x^3$$

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

At a maximum,

$$30x - 12x^2 = 0$$

$$6x(5 - 2x) = 0$$

The useful value is

$$x = 2.5$$

Now substitute into the volume:

$$V = (2.5)^2(15 - 4(2.5))$$

$$V = 6.25 \times 5$$

$$V = 31.25$$

FINAL ANSWER

31.25 cm^3

PAST PAPER QUESTION **Q#: 22** **Marks: 5**

TOPIC: **Differentiation**

Jun 2025 P2HR - Jun 2025 | P2HR

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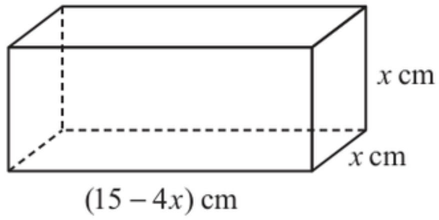

Diagram **NOT** accurately drawn

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Find the maximum value of V

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Differentiation**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

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FINAL ANSWER

31.25 cm^3

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2HR - Jun 2025 | P2HR

23 The diagram shows a triangular prism, *ABCDEF*, with a horizontal rectangular base *ABCD*

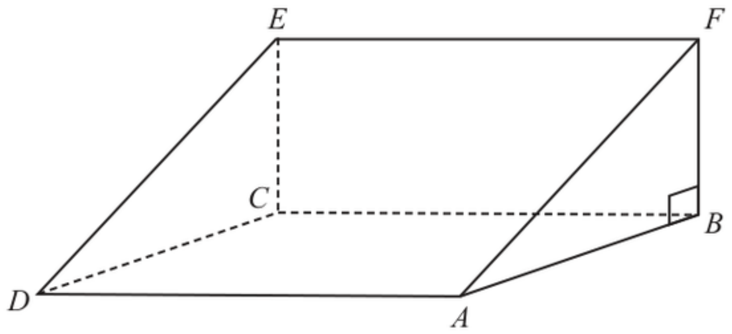

Diagram **NOT** accurately drawn

M is the midpoint of the line *AC*

$AC = 40$ cm angle $CAE = 35°$ angle $ABF = 90°$

Work out the size of angle *CME*
Give your answer correct to 3 significant figures.

.... °

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

$$AC = 40$$

The point E is vertically above C, so triangle ACE is right-angled at C.

$$\angle CAE = 35^\circ$$

$$\tan 35^\circ = \frac{CE}{AC}$$

$$CE = 40\tan 35^\circ$$

Since M is the midpoint of AC,

$$CM = 20$$

In right triangle CME,

$$\tan\angle CME = \frac{CE}{CM}$$

$$\tan\angle CME = \frac{40\tan 35^\circ}{20}$$

$$\tan\angle CME = 2\tan 35^\circ$$

$$\angle CME = 54.5^\circ$$

FINAL ANSWER

54.5°

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2HR - Jun 2025 | P2HR

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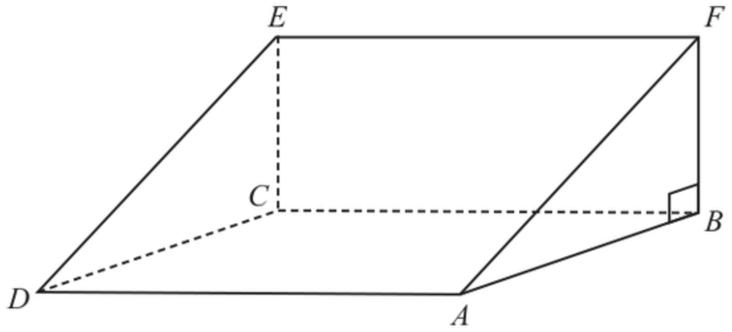

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Work out the size of angle CME
Give your answer correct to 3 significant figures.

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(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

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In right triangle CME,

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$$\tan \angle CME = \frac{40 \tan 35^\circ}{20}$$

$$\tan \angle CME = 2 \tan 35^\circ$$

$$\angle CME = 54.5^\circ$$

FINAL ANSWER

54.5°

PAST PAPER QUESTION Q#: 24 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2025 P2HR - Jun 2025 | P2HR

24 PQ is a straight line drawn on a square grid, with a scale of 1 cm for 1 unit on each axis.

P has coordinates $(-5,\ a)$ and Q has coordinates $(7,\ 3a)$ where $a > 0$

The length of PQ is $4\sqrt{10}$ cm

Find an equation of the perpendicular bisector of PQ
Give your answer in the form $y = mx + c$ where m and c are integers.

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Coordinate Geometry.

Method

$$P = (-5, a), \qquad Q = (7, 3a)$$

The length of PQ is $4\sqrt{10}$, so

$$\sqrt{(7-(-5))^2 + (3a-a)^2} = 4\sqrt{10}$$

$$\sqrt{12^2 + (2a)^2} = 4\sqrt{10}$$

Square both sides:

$$144 + 4a^2 = 160$$

$$4a^2 = 16$$

$$a^2 = 4$$

Since $a > 0$,

$$a = 2$$

So

$$P = (-5, 2), \qquad Q = (7, 6)$$

The midpoint of PQ is

$$\left(\frac{-5+7}{2}, \frac{2+6}{2}\right) = (1, 4)$$

The gradient of PQ is

$$\frac{6-2}{7-(-5)} = \frac{4}{12} = \frac{1}{3}$$

So the gradient of the perpendicular bisector is

$$-3$$

Using point $(1, 4)$:

$$y - 4 = -3(x - 1)$$

$$y = -3x + 7$$

FINAL ANSWER

$\boxed{y = -3x + 7}$

PAST PAPER QUESTION

Q#: 24 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2025 P2HR - Jun 2025 | P2HR

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PAST PAPER SOLUTION Q#: 24 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

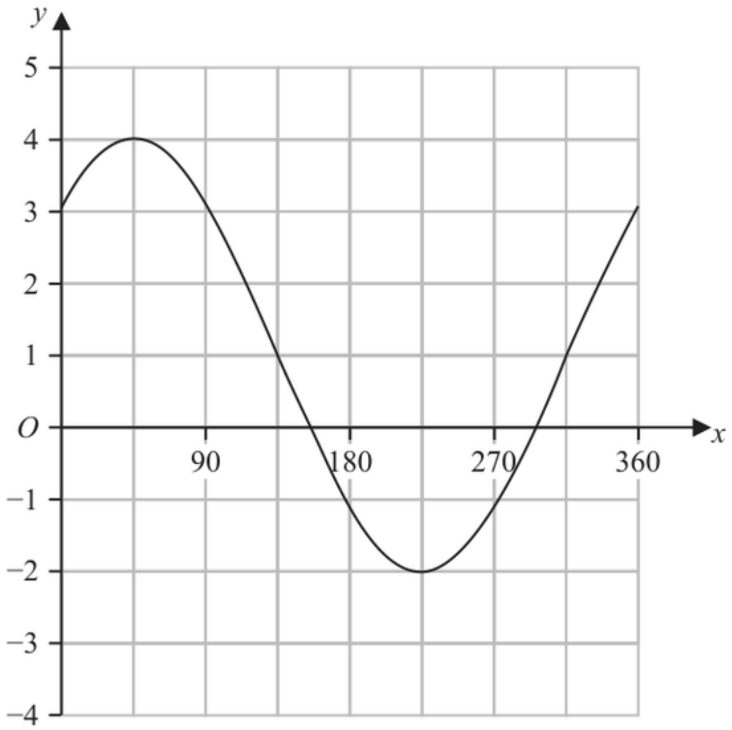
$\boxed{y = -3x + 7}$

PAST PAPER QUESTION Q#: 25 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2025 P2HR - Jun 2025 | P2HR

25 The graph of $y = a\sin(x + b)^{\circ} + c$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Find a suitable value for a, for b and for c

a = ...

b = ...

c = ...

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

From the graph,

$$\text{maximum} = 4$$

$$\text{minimum} = -2$$

The amplitude is

$$a = \frac{4 - (-2)}{2} = 3$$

The centre line is

$$c = \frac{4 + (-2)}{2} = 1$$

The maximum is about $45°$. For a sine graph, a maximum happens when the angle inside the sine is $90°$, so

$$45 + b = 90$$

$$b = 45$$

FINAL ANSWER

$a = 3, \ b = 45, \ c = 1$

PAST PAPER QUESTION Q#: 25 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2025 P2HR - Jun 2025 | P2HR

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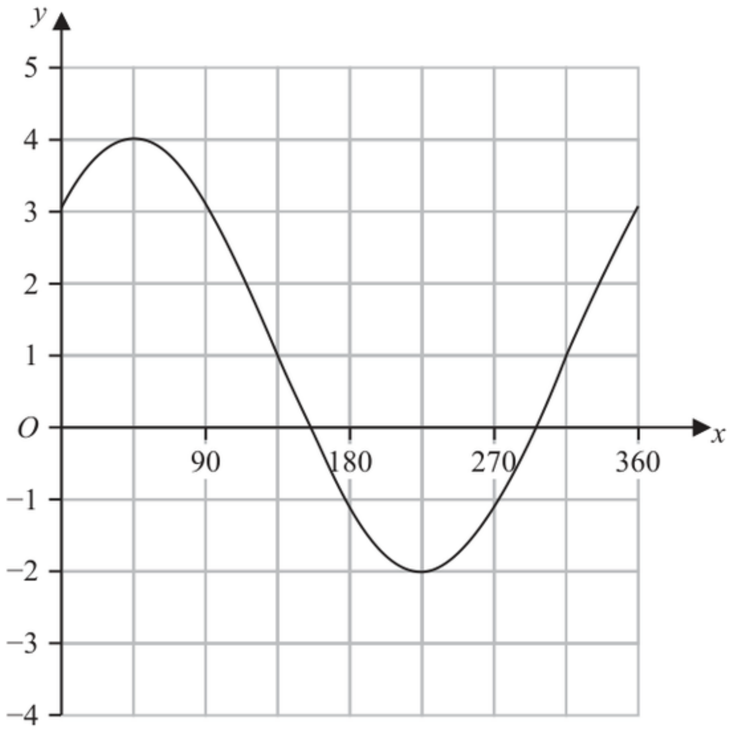

Find a suitable value for a, for b and for c

$a =$ ....

$b =$ ....

$c =$ ....

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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From the graph,

$$\text{maximum} = 4$$

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The amplitude is

$$a = \frac{4 - (-2)}{2} = 3$$

The centre line is

$$c = \frac{4 + (-2)}{2} = 1$$

The maximum is about 45°. For a sine graph, a maximum happens when the angle inside the sine is 90°, so

$$45 + b = 90$$

$$b = 45$$

FINAL ANSWER

$a = 3,\ b = 45,\ c = 1$

PAST PAPER QUESTION Q#: 26 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2025 P2HR - Jun 2025 | P2HR

26 The diagram shows a solid hemisphere, H

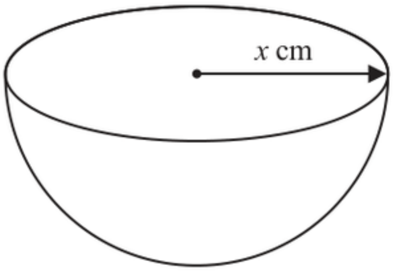

Diagram **NOT** accurately drawn

The radius of H is x cm
The volume of H is 6174π cm³

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm

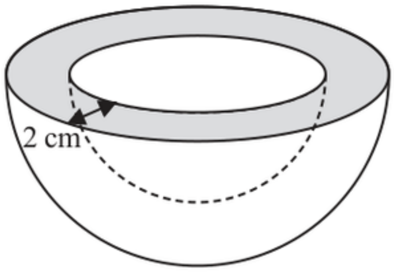

Diagram **NOT** accurately drawn

Work out the **total** surface area of the bowl.
Give your answer in terms of π

.... cm²

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The volume of a hemisphere is

$$\frac{2}{3}\pi r^3$$

For the original hemisphere H,

$$\frac{2}{3}\pi x^3 = 6174\pi$$

Cancel π:

$$\frac{2}{3}x^3 = 6174$$

$$x^3 = 9261$$

$$x = 21$$

The outer radius of the bowl is 21 cm.
The thickness is 2 cm, so the inner radius is

$$21 - 2 = 19$$

Total surface area of the bowl is:

- outer curved hemisphere
- inner curved hemisphere
- flat annular rim

Outer curved surface area:

$$2\pi(21)^2 = 882\pi$$

Inner curved surface area:

$$2\pi(19)^2 = 722\pi$$

Flat rim area:

$$\pi(21^2 - 19^2) = 80\pi$$

Total:

$$882\pi + 722\pi + 80\pi = 1684\pi$$

FINAL ANSWER

$\boxed{1684\pi\ \text{cm}^2}$

PAST PAPER QUESTION Q#: 26 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2025 P2HR - Jun 2025 | P2HR

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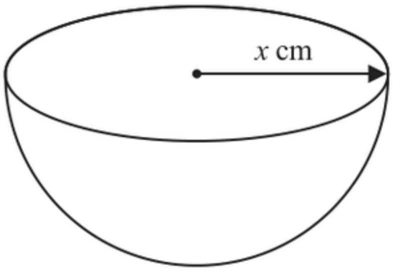

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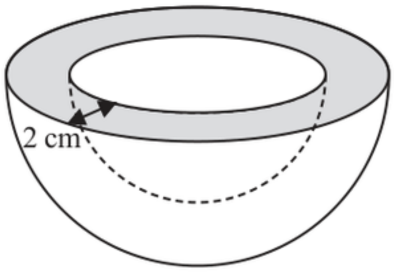

Diagram **NOT** accurately drawn

Work out the **total** surface area of the bowl.
Give your answer in terms of π

.... cm^2

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2025 P2HR - Jun 2025 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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Flat rim area:

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Total:

$$882\pi + 722\pi + 80\pi = 1684\pi$$

FINAL ANSWER

$\boxed{1684\pi\ \text{cm}^2}$